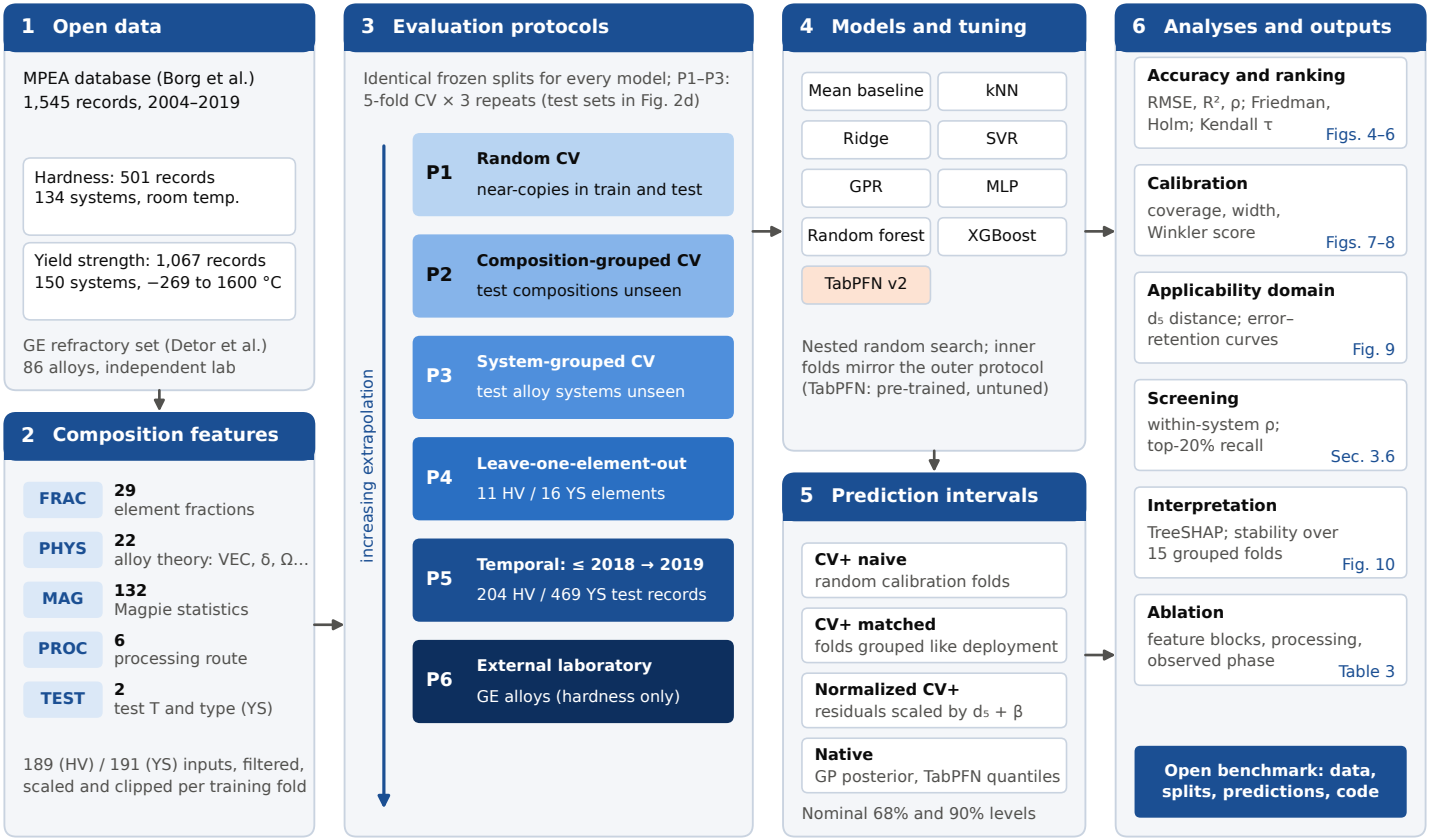
1 Open data
MPEA database (Borg et al.)
1,545 records, 2004–2019
Hardness: 501 records
134 systems, room temp.
Yield strength: 1,067 records
150 systems, −269 to 1600 °C
GE refractory set (Detor et al.)
86 alloys, independent lab
2 Composition features
FRAC 29 element fractions
PHYS 22 alloy theory: VEC, δ, Ω...
MAG 132 Magpie statistics
PROC 6 processing route
TEST 2 test T and type (YS)
189 (HV) / 191 (YS) inputs, filtered, scaled and clipped per training fold
3 Evaluation protocols
Identical frozen splits for every model; P1–P3: 5-fold CV × 3 repeats (test sets in Fig. 2d)
increasing extrapolation
P1 Random CV
near-copies in train and test
P2 Composition-grouped CV
test compositions unseen
P3 System-grouped CV
test alloy systems unseen
P4 Leave-one-element-out
11 HV / 16 YS elements
P5 Temporal: ≤ 2018 → 2019
204 HV / 469 YS test records
P6 External laboratory
GE alloys (hardness only)
4 Models and tuning
Mean baseline
kNN
Ridge
SVR
GPR
MLP
Random forest
XGBoost
TabPFN v2
Nested random search; inner folds mirror the outer protocol (TabPFN: pre-trained, untuned)
5 Prediction intervals
CV+ naive
random calibration folds
CV+ matched
folds grouped like deployment
Normalized CV+
residuals scaled by d_s + β
Native
GP posterior, TabPFN quantiles
Nominal 68% and 90% levels
6 Analyses and outputs
Accuracy and ranking
RMSE, R^2, ρ; Friedman, Holm; Kendall τ
Figs. 4–6
Calibration
coverage, width, Winkler score
Figs. 7–8
Applicability domain
d_s distance; error–retention curves
Fig. 9
Screening
within-system ρ; top-20% recall
Sec. 3.6
Interpretation
TreeSHAP; stability over 15 grouped folds
Fig. 10
Ablation
feature blocks, processing, observed phase
Table 3
Open benchmark: data, splits, predictions, code